Visual Autoregressive (VAR) and Diffusion Models for Inverse Design of Metamaterials

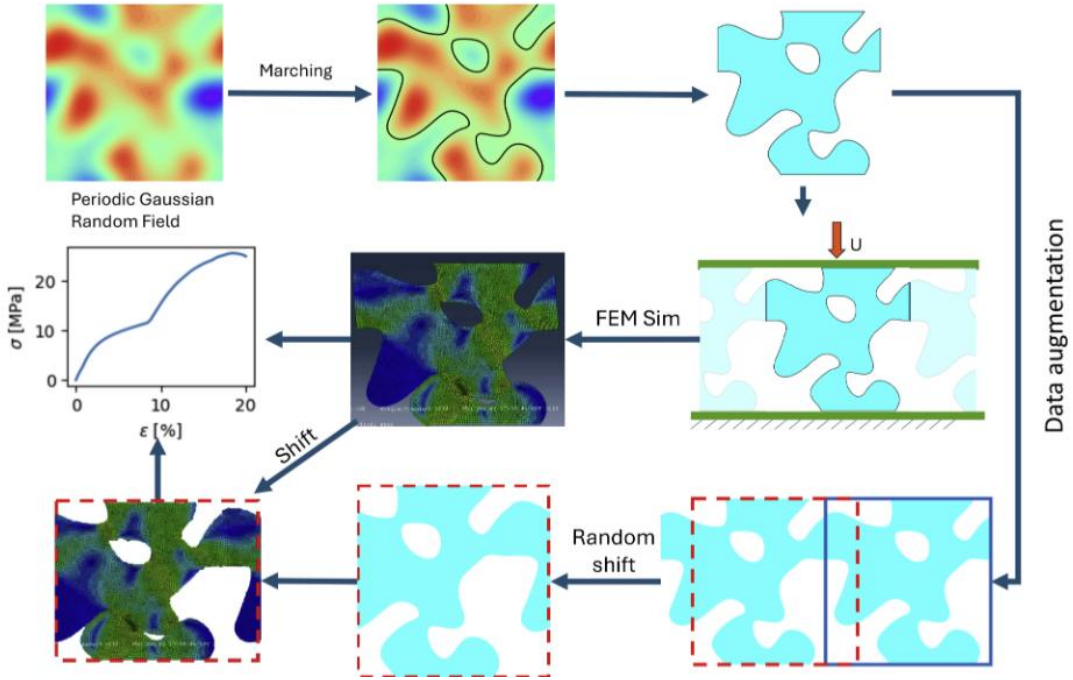
Hossein Kazemi

Department of Civil and Environmental Engineering

ILLINOIS

Statement of Problem

- The inverse design of metamaterial unit cells aims to **generate geometries that achieve a target stress–strain response**.
- Data generation process is as follows:



Diffusion Model - Intro

Diffusion models learn to generate data by three main steps:

1. **Forward process:** gradually add Gaussian noise to real data until it becomes pure noise.
2. **Reverse process:** train a neural network to remove that noise step by step.
3. **Sampling:** start from random noise and iteratively denoise into a new sample.

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon, \qquad \epsilon \sim N(0, I)$$

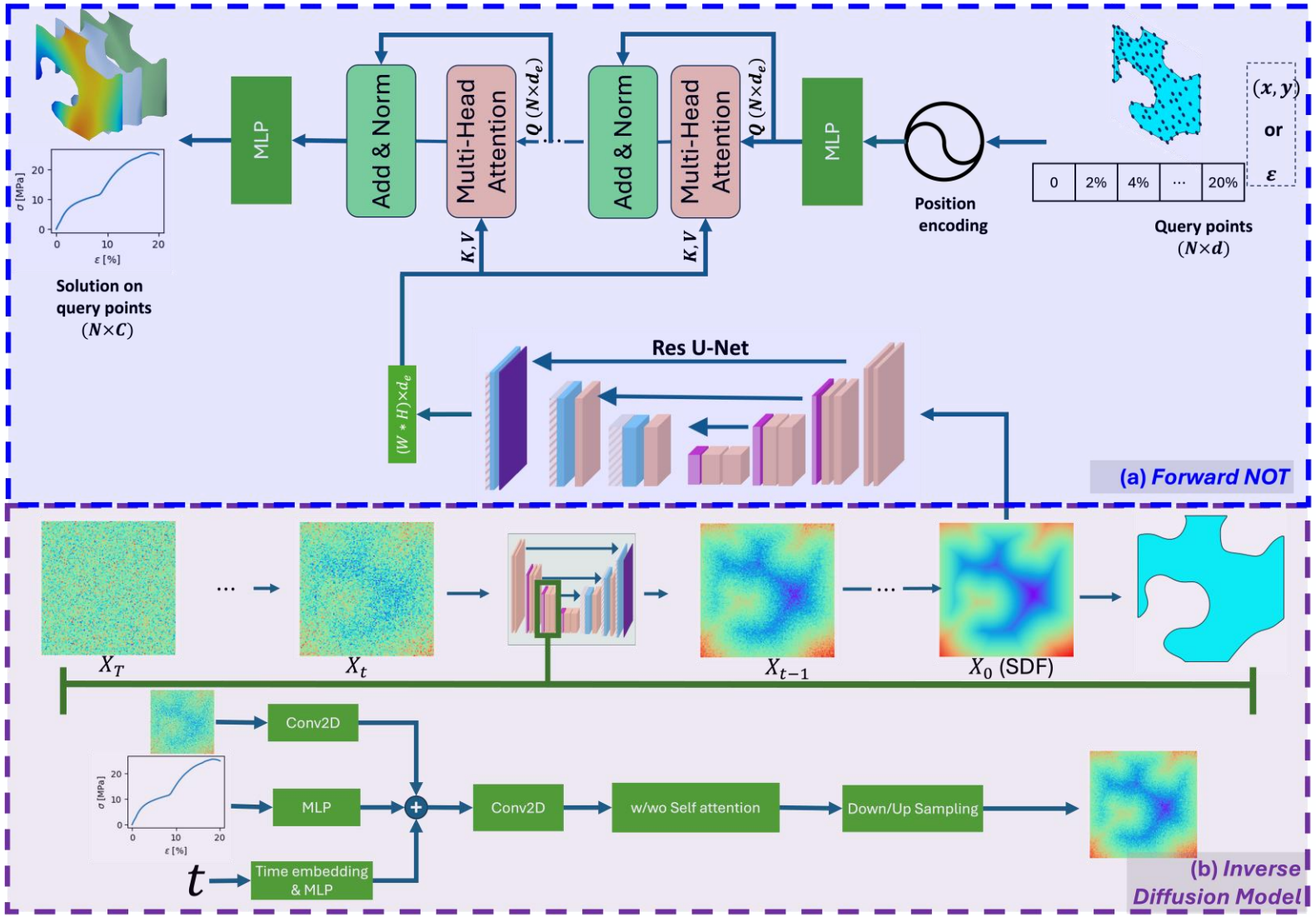
Noise at each diffusion step

$$L = E\left[\left\|\epsilon - \epsilon_{\theta(x_t, t)}\right\|^2\right]$$

Training Loss

Start with: $x_T \sim N(0, I)$ → Then repeatedly denoise: $x_t \rightarrow x_{t-1} \rightarrow x_{t-2} \rightarrow \ldots \rightarrow x_0$

Diffusion Model - Deployed Network Architecture, Loss Function



$$\mathcal{L}_{diff}(\theta) = E_{x_0,y,t,\epsilon}\left[\left\|\epsilon - \epsilon_{\theta(x_t,t,y)}\right\|_2^2\right]$$

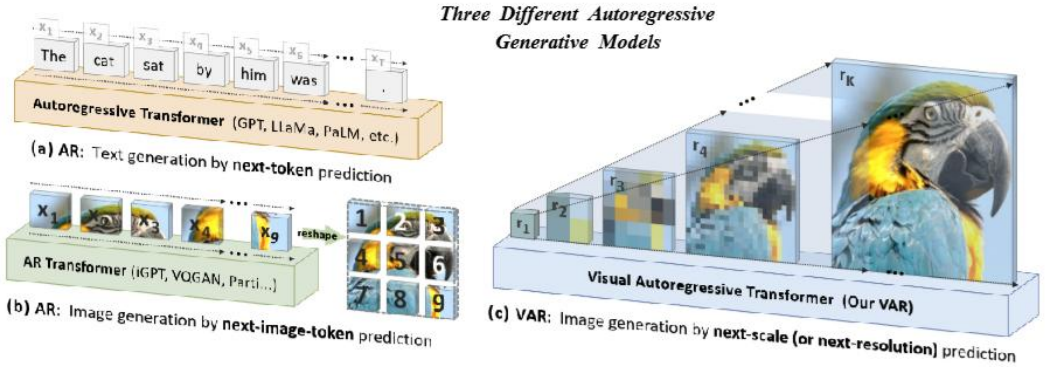
x_0: Clean normalized SDF image of the metamaterial unit cell.

x_t: Noisy SDF at timestep t

y: Target stress-strain curve used as the conditioning input.

ϵ: Gaussian noise

Visual Autoregressive (VAR) Model - Intro



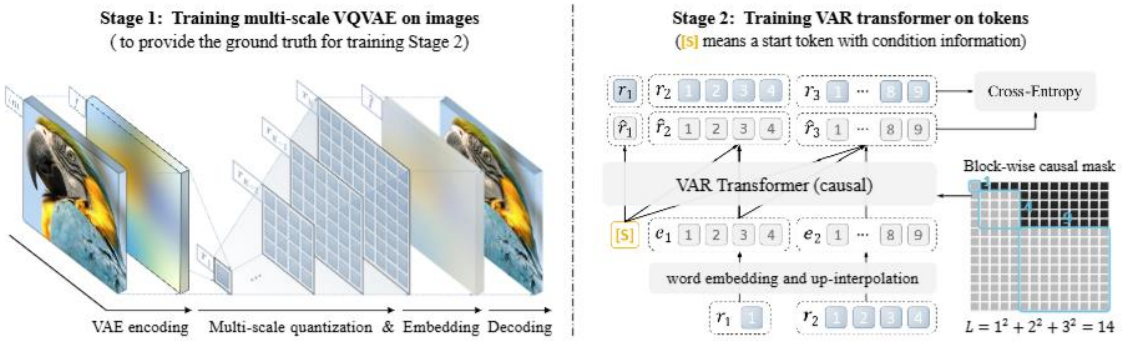
Standard AR vs VAR
(a) AR applied to language
(b) AR applied to images
(c) VAR for images

$$p(x_1, x_2, \dots, x_T) = \prod_{t=1}^{T} p\left(x_t \middle| x_1, x_2, \dots, x_{(t-1)}\right)$$

$$f = \mathcal{E}(im), \boxed{q = Q(f)}$$

$$q^{(i,j)} = (\arg min)_{v \in [V]} \left\| lookup(Z, v) - f^{(i,j)} \right\|_2 \in [V]$$

$$p(r_1, r_2, \dots, r_K) = \prod_{k=1}^{K} p\left(r_k \middle| r_1, r_2, \dots, r_{(k-1)}\right)$$



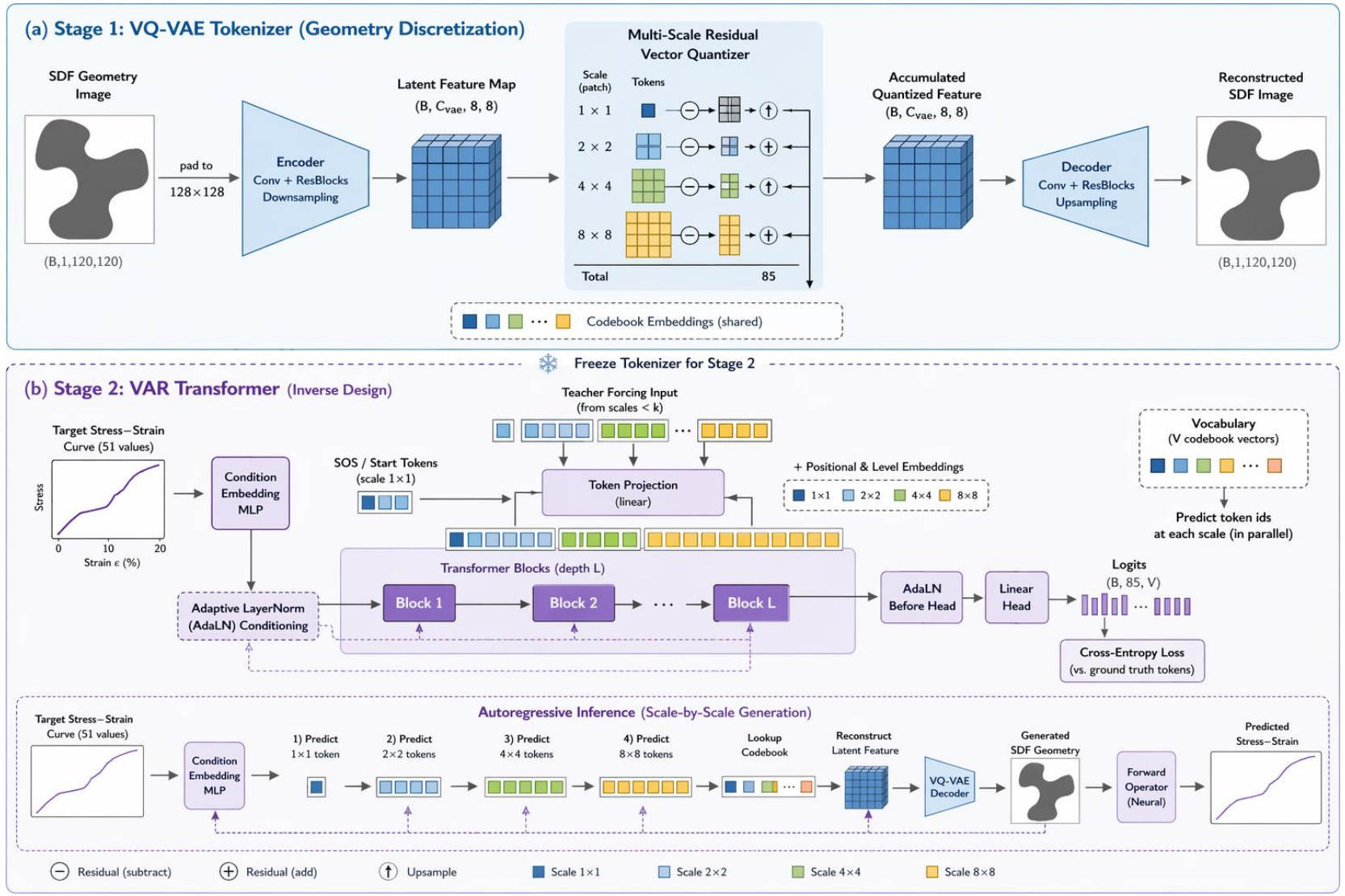
Algorithm 1: Multi-scale VQVAE Encoding

Inputs: raw image im;
Hyperparameters: steps K, resolutions $(h_k, w_k)_{k=1}^{K}$;
$f = \mathcal{E}(im)$, $R = []$;
for $k = 1, \cdots, K$ **do**
  $r_k = \mathcal{Q}(\text{interpolate}(f, h_k, w_k))$;
  $R = \text{queue_push}(R, r_k)$;
  $z_k = \text{lookup}(Z, r_k)$;
  $z_k = \text{interpolate}(z_k, h_K, w_K)$;
  $f = f - \phi_k(z_k)$;
Return: multi-scale tokens R;

Algorithm 2: Multi-scale VQVAE Reconstruction

Inputs: multi-scale token maps R;
Hyperparameters: steps K, resolutions $(h_k, w_k)_{k=1}^{K}$;
$\hat{f} = 0$;
for $k = 1, \cdots, K$ **do**
  $r_k = \text{queue_pop}(R)$;
  $z_k = \text{lookup}(Z, r_k)$;
  $z_k = \text{interpolate}(z_k, h_K, w_K)$;
  $\hat{f} = \hat{f} + \phi_k(z_k)$;
$\hat{im} = \mathcal{D}(\hat{f})$;
Return: reconstructed image $\hat{im}$;

VQVAE and VAR Models – Deployed Network Architectures, Loss Functions



- In the VQVAE stage, the codebook tokens are trained such that some of them represent coarse image information, and some finer details
- In VAR stage, we train a Transformer model to predict tokens that represent resolutions in each step

$$\mathcal{L}_{\mathrm{VAR}}(\theta) = -\left(\frac{1}{L}\right)\sum_{i=1}^{L} \log\left(p_\theta\left(z_i \mid z_{(<i)}, y\right)\right)$$

L: Total number of tokens in one training sample

z_i: The token at position i

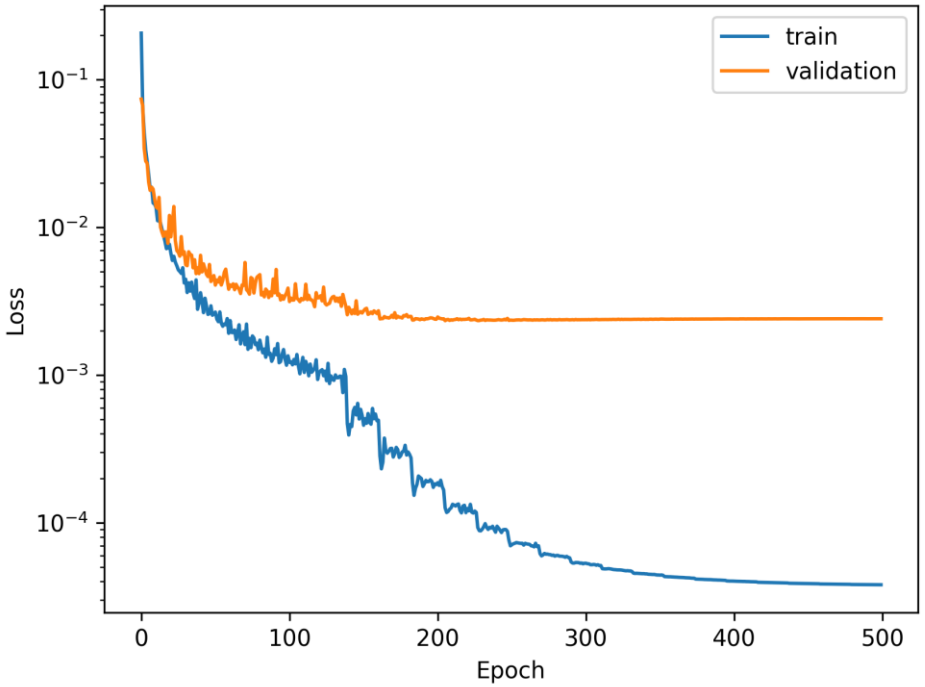
Reconstruction likelihood

$$\mathcal{L}_{VQVAE}(\theta) = \log p\left(x \middle| z_{q(x)}\right) + \left\|sg\left[z_{e(x)}\right] - e\right\|_2^2 + \beta\left\|z_{e(x)} - sg[e]\right\|_2^2$$

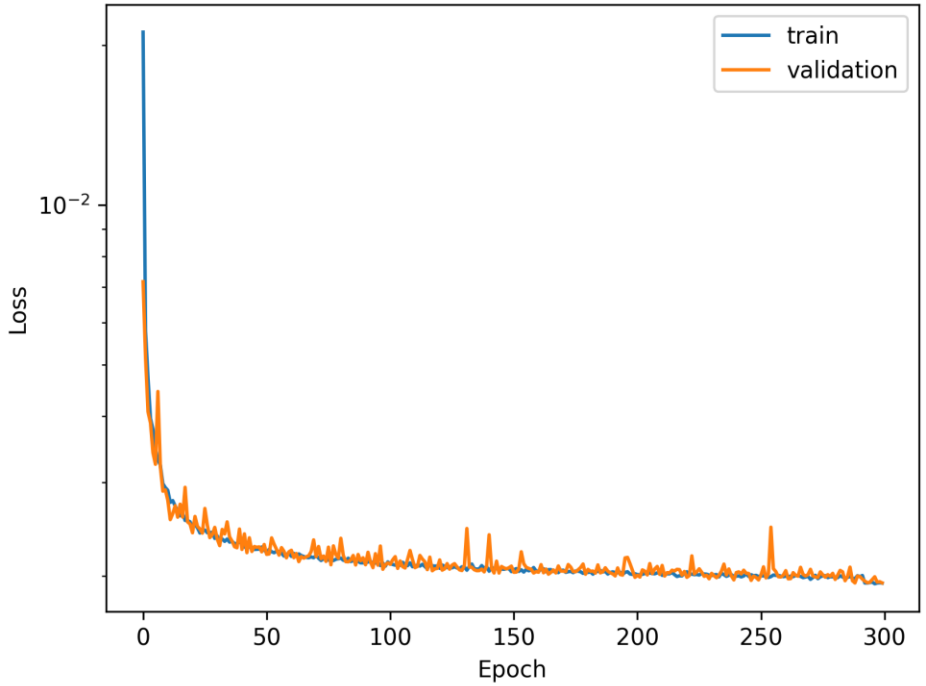
Codebook loss

Forces encoder outputs to stay close to selected embedding

Results – Training History

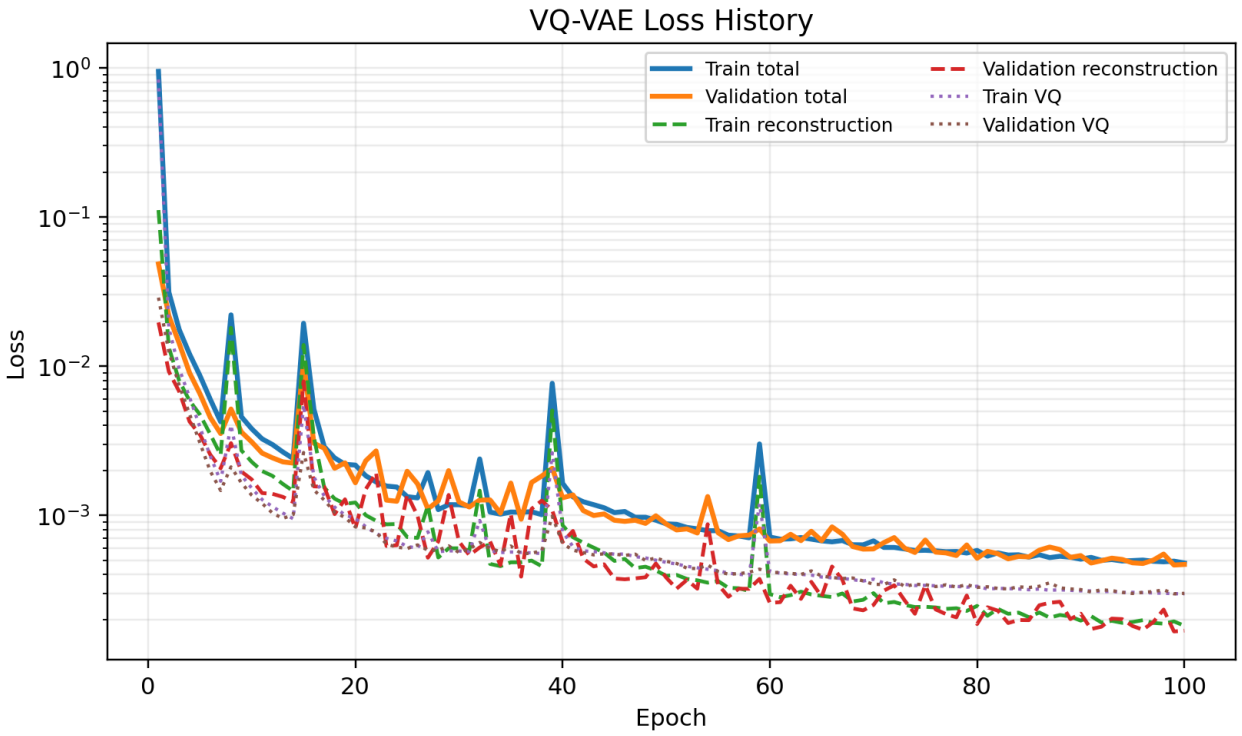


Attention-based Neural Operator Training

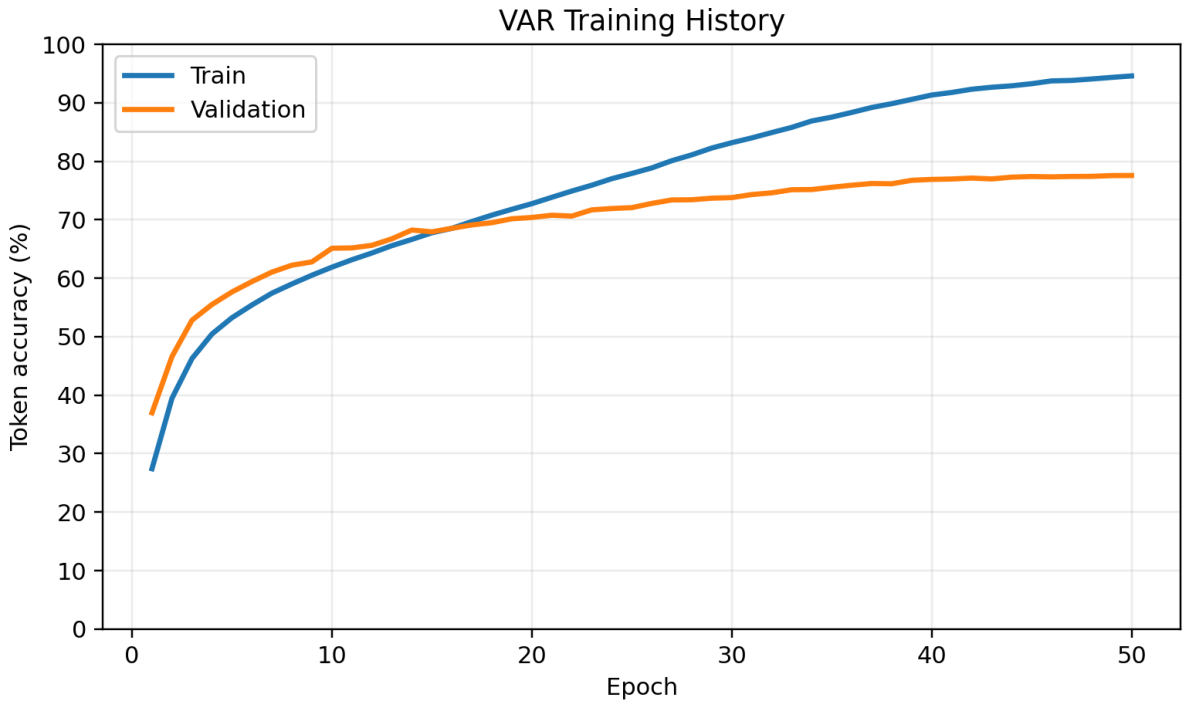


Diffusion Training

Results – Training History

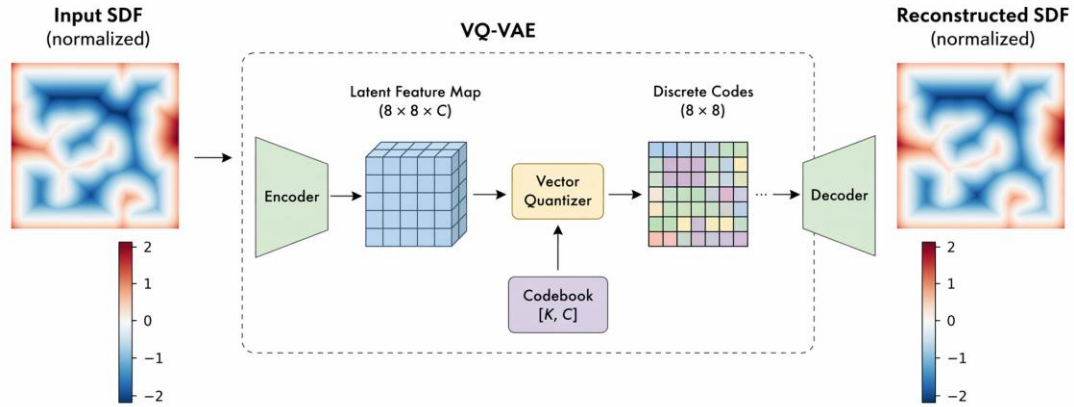


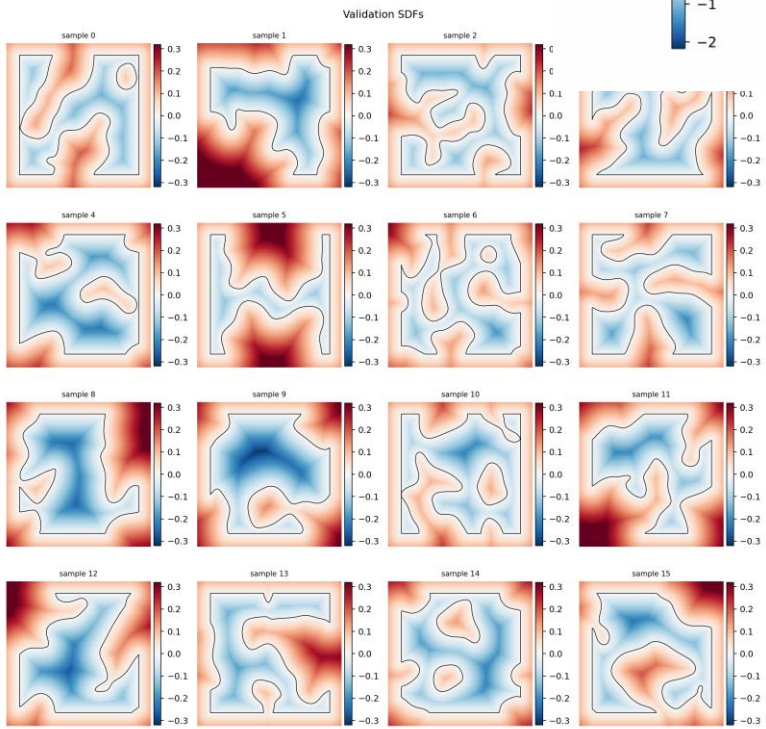
VQVAE

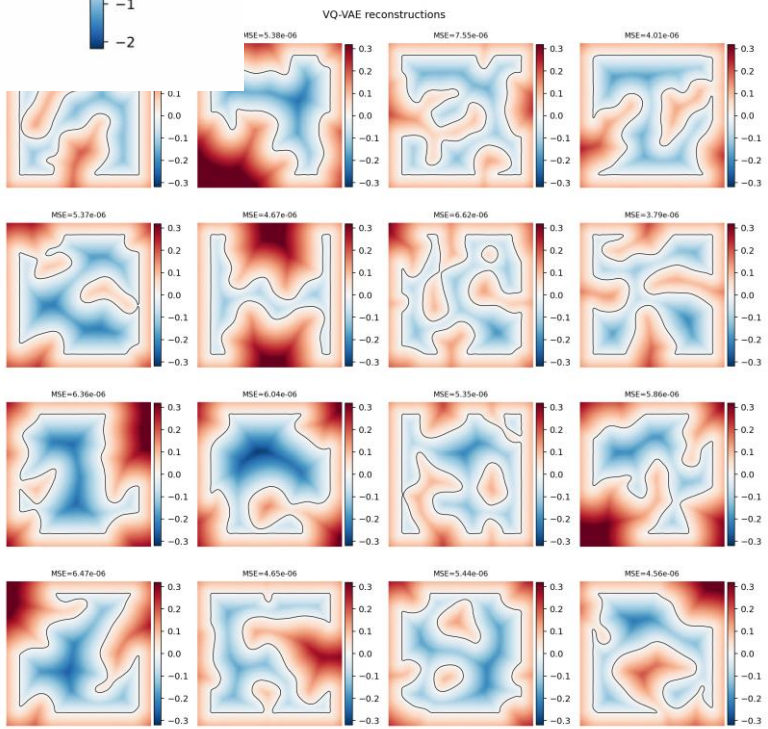


VAR

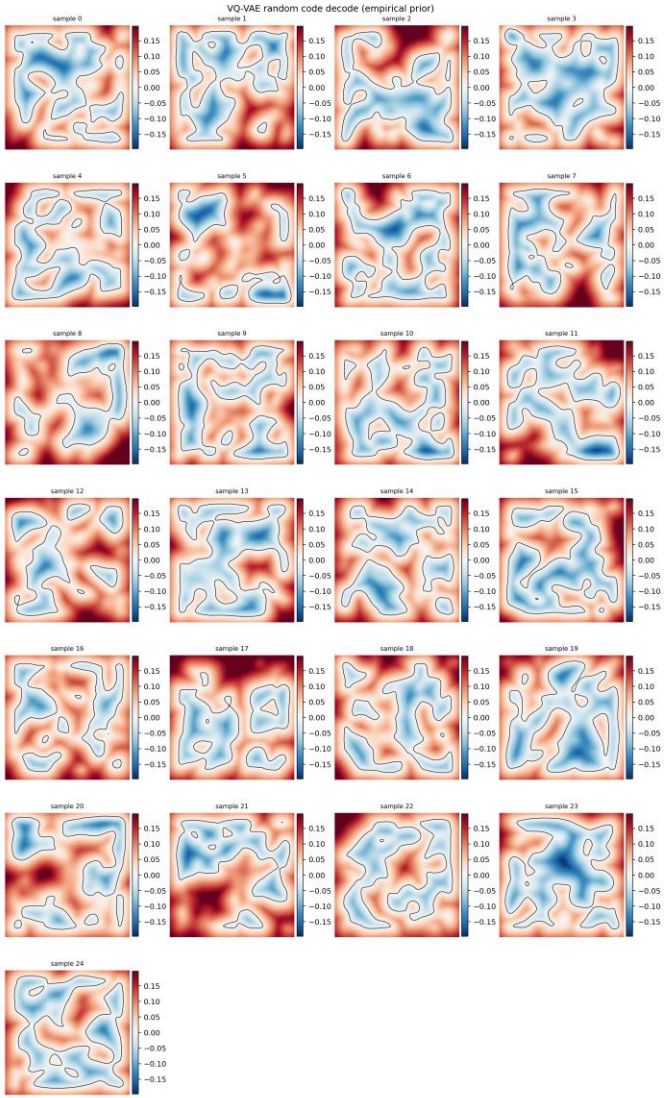
Results – VQVAE Reconstruction and VAR Generation





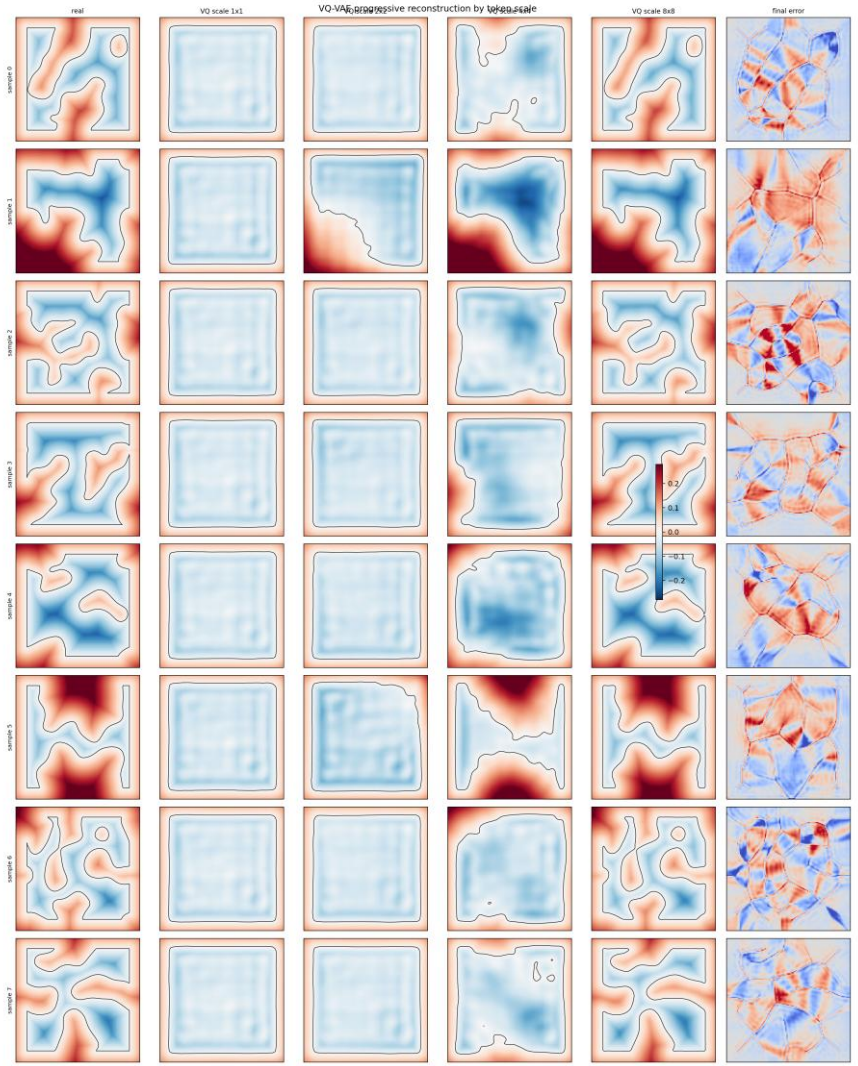


VQVAE Progressive Reconstruction by token scale

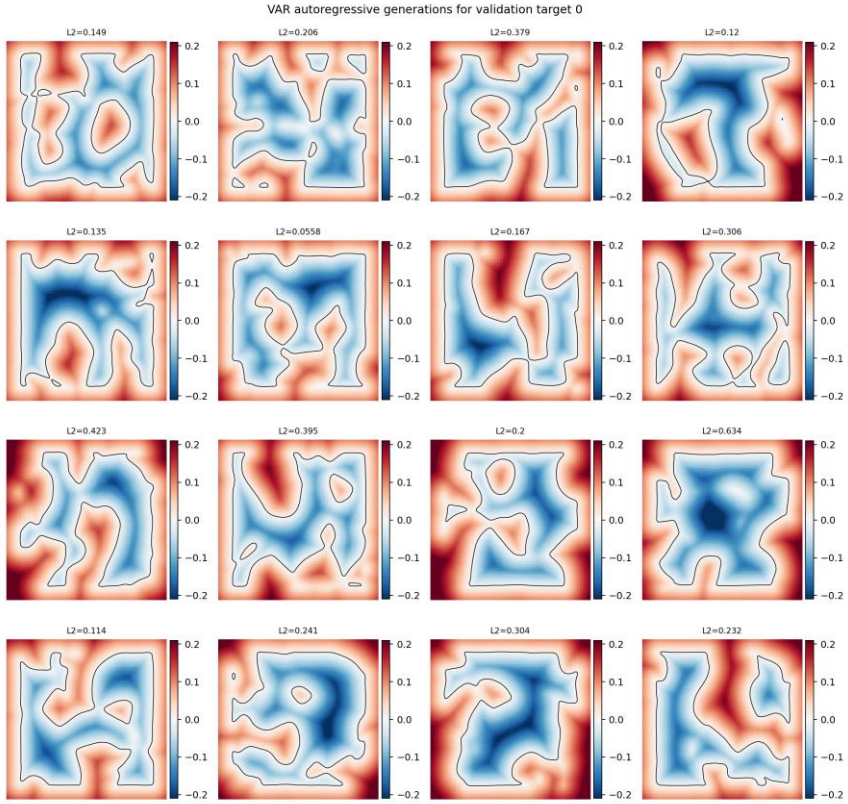


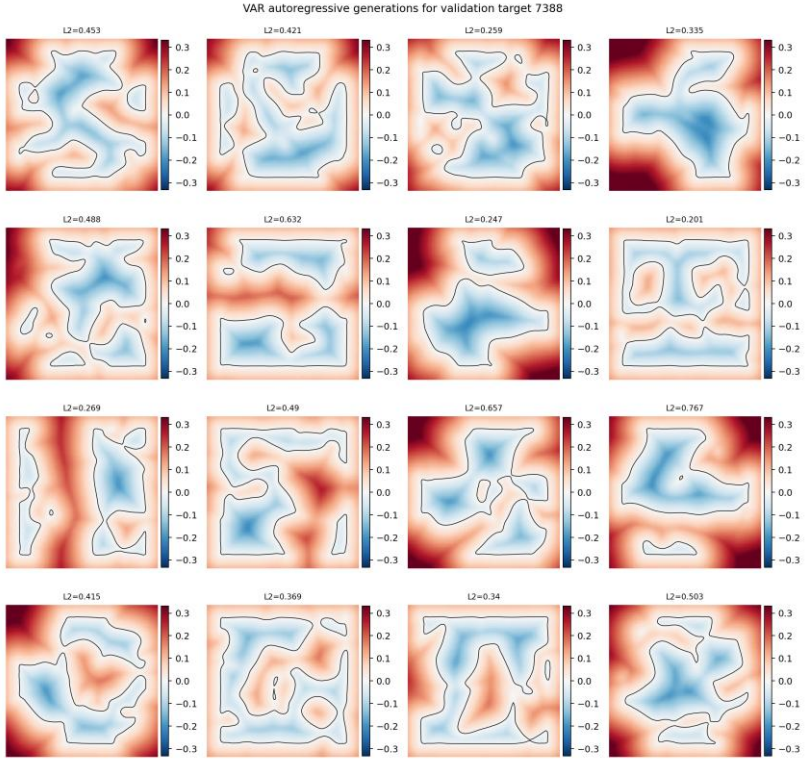
What do these discrete tokens mean?

VQVAE progressive reconstruction



Results – VQVAE Reconstruction and VAR Generation

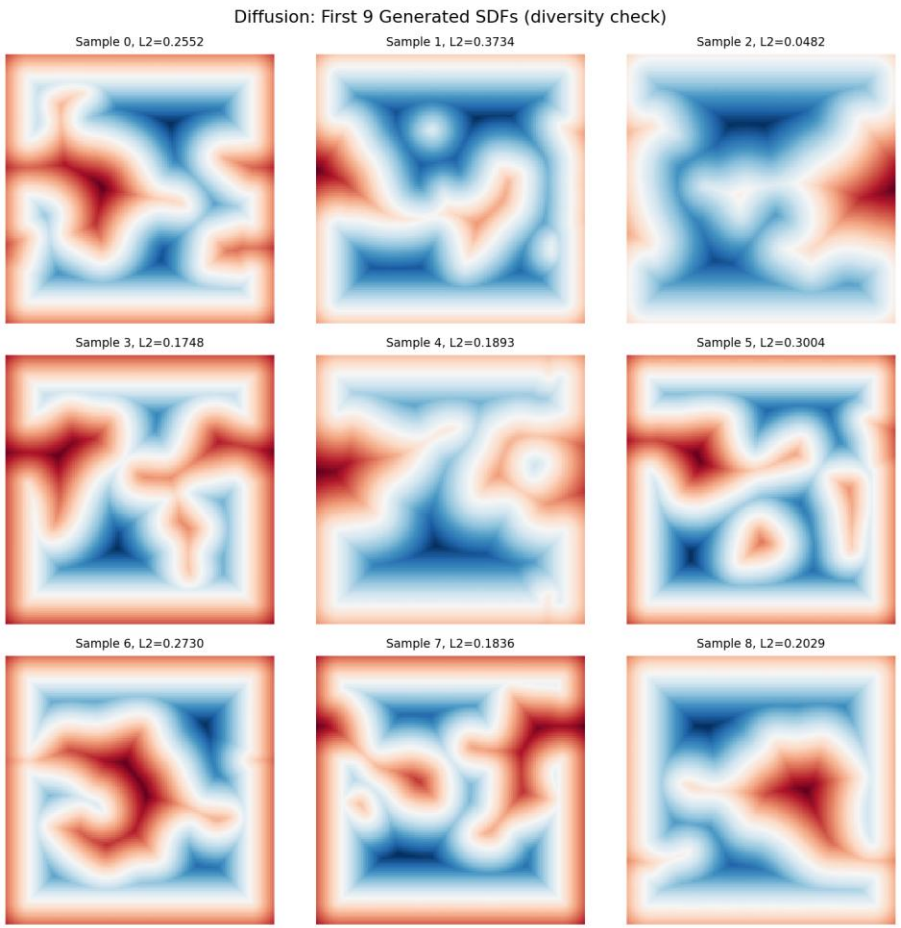


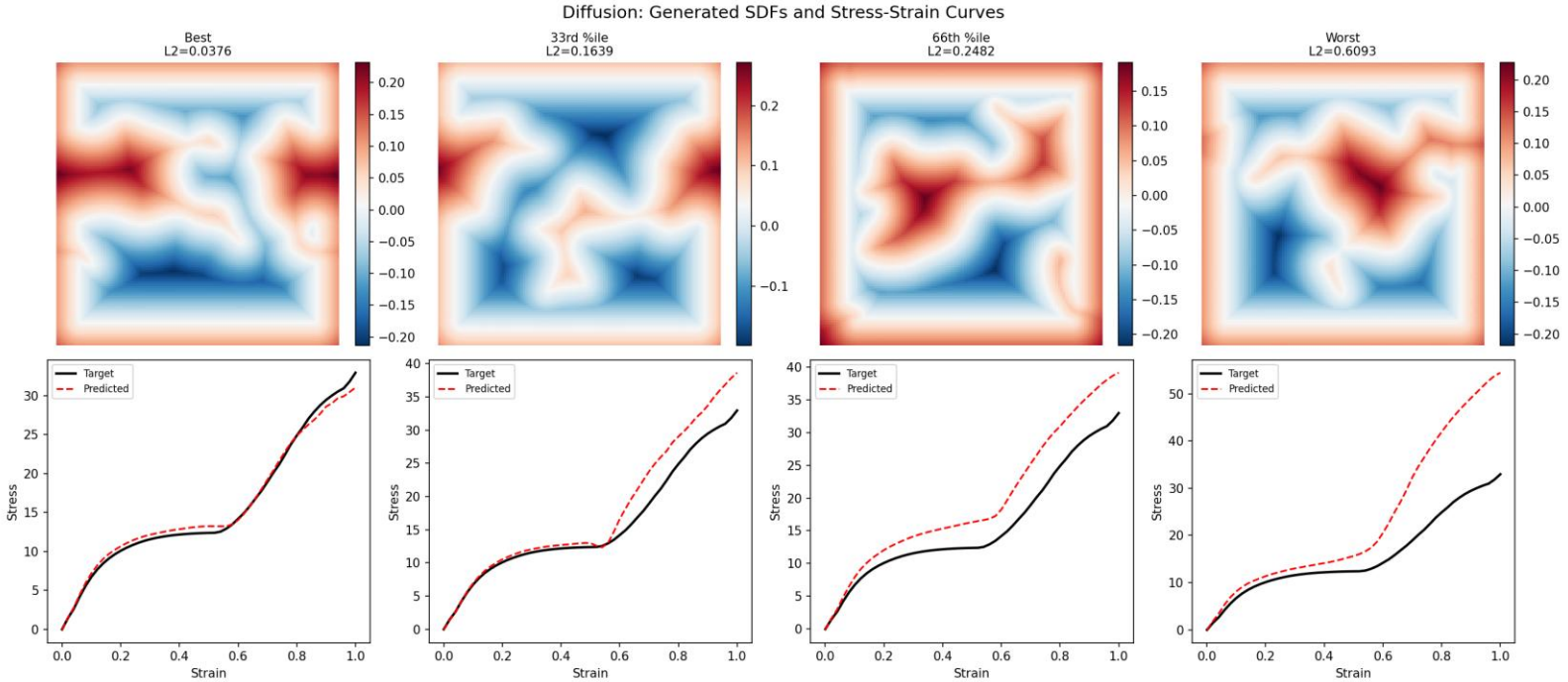


Problem:

- Here, VAR starts from <sos> token **OR** SS curve
- The reconstruction quality seems acceptable; however, the samples that are generated from VAR are either blurry images or have island materials
- 1) Discrete representation here is hurting the performance **OR** 2) We can have many tokens variations that leads to the same SS curve; so the VAR model is confused during training because of the same input but different outputs **OR** 3) Maybe changing the network architecture can improve the results

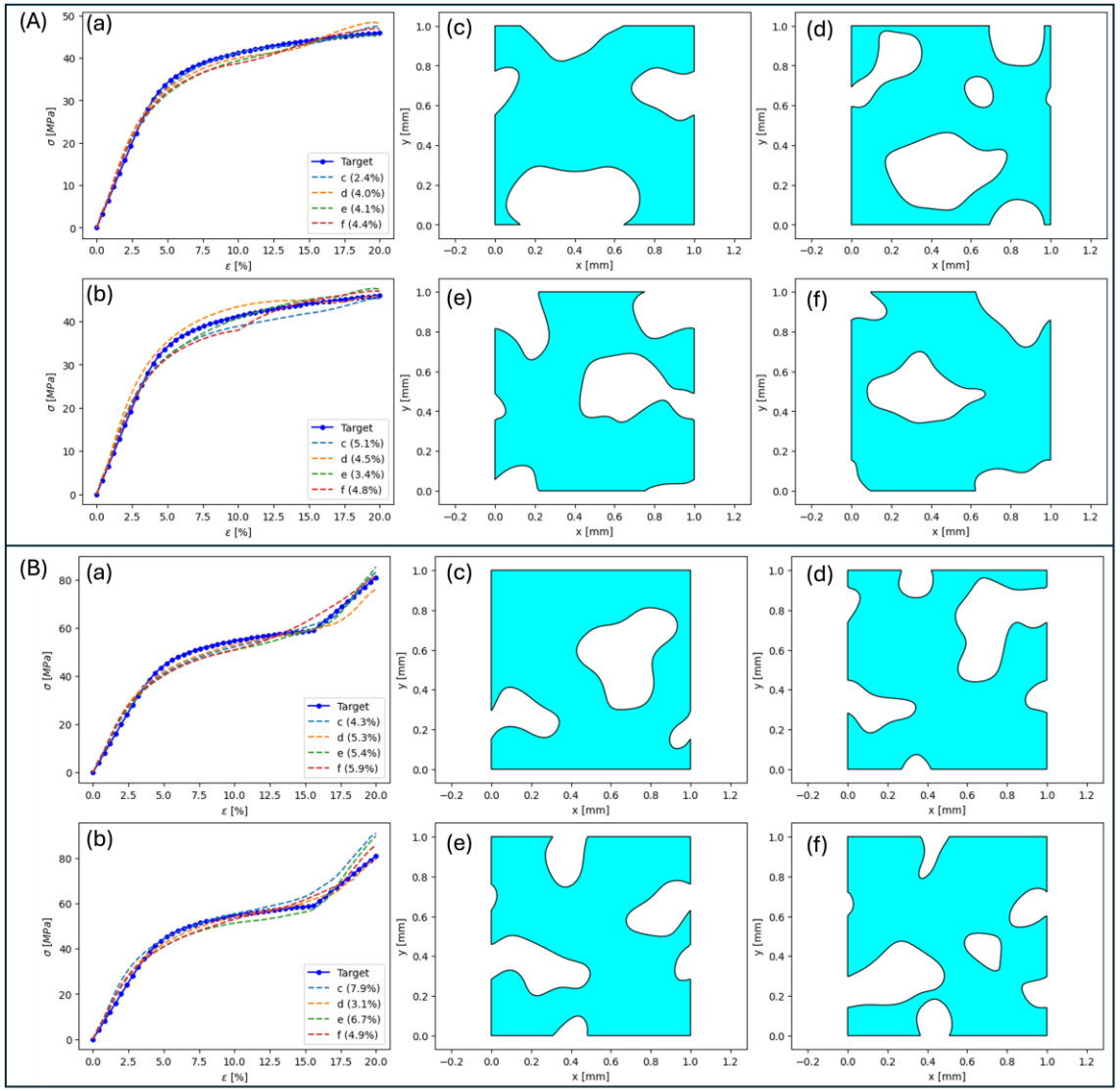
Results – Diffusion Model Metamaterial Generation





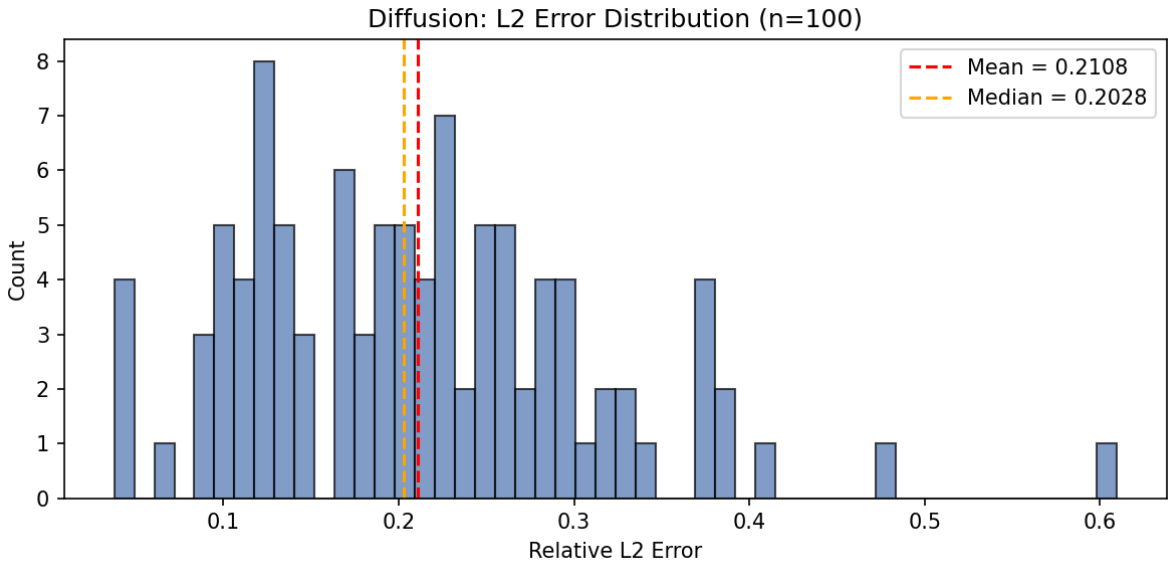
- Samples seem more realistic than VAR generated samples
- + Average L2 error is lower than VAR (0.21 vs 0.29)

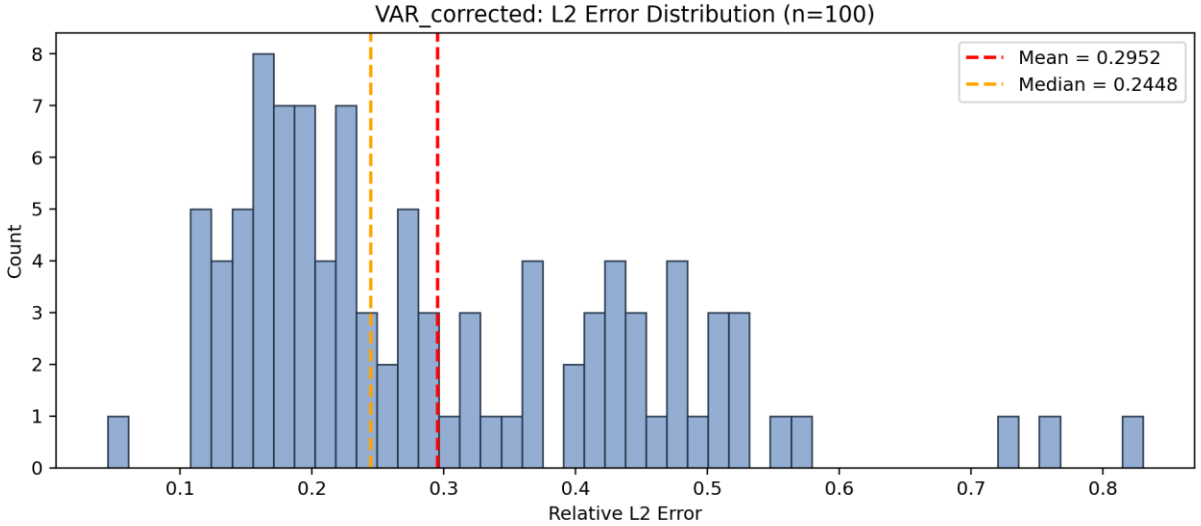
Results – Diffusion Generated Samples



- These two target curves are fed into the inverse diffusion model for 500 SDF solutions
- Then fed into the forward models (an attention-based neural operator) for prediction
- We select the four design results with the lowest L2 relative error, shown in (c–f)
- (a) → Comparison between the target and prediction of (c–f) using the forward model
- (b) → Corresponding stress-strain curves obtained using Abaqus

Results – VAR and Diffusion Final Comparison





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Thank You for Your Attention!

mkazemi3@illinois.edu